

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed embedded C/C++ firmware for ESP32 microcontrollers, implementing **Bluetooth Low Energy (BLE)** and **Wi-Fi connectivity** for smart home devices, achieving a **99.8% connection success rate**.
- Designed and integrated communication stacks, including **Modbus RTU over RS-485** and **Modbus TCP**, enabling reliable data exchange with energy meters and inverters, reducing integration time by **20%**.
- Implemented **RTOS-based development** with FreeRTOS for real-time task management on ESP32, optimizing memory usage by **15%** in constrained environments.
- Authored Python scripts for automated testing of firmware builds and device provisioning, reducing manual testing cycles by **40%**.
- Integrated device logic with cloud IoT infrastructure using **MQTT** and **REST APIs**, implementing secure device provisioning and **OTA update workflows**.
- Enhanced embedded system security by implementing **TLS authentication** and secure coding practices, mitigating potential vulnerabilities in production firmware.

SKILLS

Embedded Systems – Embedded C/C++, ESP32, ESP-IDF, FreeRTOS, Microcontrollers, Bare-metal development, Memory Management, RTOS

Wireless & Wired Protocols – Bluetooth Low Energy (BLE), Wi-Fi, Modbus RTU, Modbus TCP, RS-485, I2C, UART

Cloud & IoT – MQTT, REST APIs, WebSockets, Protobuf, AWS IoT Core, Device Provisioning, OTA Updates

Scripting & Automation – Python, Shell Scripting, Jenkins, Git

Linux & Tools – Linux Environments, Cross-compiling Toolchains, Serial/USB Interfaces, Docker

Security – Public-Key Cryptography, TLS Authentication, Secure Boot, Secure Coding Practices

Data & Analytics – Power BI, DAX, Power Query, Tableau, R, Python (pandas, NumPy, scikit-learn)

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Smart Home Energy Management Gateway (HCL Technologies)

- Developed firmware for a central gateway device using **Embedded C++** on an **ESP32** platform, managing connectivity for multiple smart home devices via **BLE and Wi-Fi**.
- Implemented **Modbus TCP** communication for energy meters and **MQTT** for cloud data ingestion, ensuring seamless data flow and achieving **99.95% data availability**.

- Integrated secure device onboarding using **certificate-based authentication** and implemented **OTA firmware updates**, reducing deployment risks and ensuring timely patches across the device fleet.

Power BI Financial Operations Dashboard (MSc Project)

- Designed and built an interactive **Power BI dashboard** to visualize key performance indicators for a financial services operations dashboard, consolidating data from multiple sources.
- Developed complex **DAX measures** and performed data modeling to support detailed transaction monitoring and reconciliation workflow analysis, enabling stakeholders to identify anomalies **30% faster**.
- Automated data extraction and transformation using **Power Query**, reducing manual data preparation time by **75%** and ensuring data accuracy for reporting.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)